METRIC

A-A-51461B
30 January 1989
SUPERSEDING
A-A-51461A
14 September 1987

COMMERCIAL ITEM DESCRIPTION

TEST KIT, TEST STRIPS AND COLOR CHART, ANTIFREEZE, FREEZE POINT AND CORROSION

This Commercial Item Description covers a test strip suitable for determining both the freezing point and reserved alkalinity of ethylene glycol antifreeze solutions used in engine cooling systems.

Salient characteristics:

The test kit shall be a test strip suitable for determining both the freezing point and reserved alkalinity (corrosion inhibition) of ethylene glycol antifreeze solutions used in engine cooling systems.

The test kit shall be a plastic strip with two impregnated reagent pads at the distal end.

The test kit shall be supplied with instructions, and color chart(s) for interpreting the color reactions on each reagent pad as a result of testing a cooling solution in accordance with the manufacturer's instructions.

The test kit shall indicate the freezing point and corrosion inhibition of an antifreeze solution by dipping the distal end into the solution and then matching the reaction on each reagent pad with the appropriate color chart.

The test kit shall have one reagent pad on the test strip that reacts to various concentrations of alkalinity in cooling solutions by changing to the following colors.

a. Blue to indicate that solution has a "GOOD" condition for corrosion protection.
b. Darker green to indicate that solution has a "BORDERLINE" condition for corrosion protection.
c. Yellow-green to indicate that solution has a "POOR" condition for corrosion protection.

AMSC N/A FSC 6630

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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The manufacturer's product shall conform to the prescribed color change reactions for reserve alkalinity (as specified within this document) when tested with the appropriate standard solutions (see Reference Standards Table and Reactions).

Reference Standards Table and Reactions

Reference Standards	Color	Condition
*Solution A	Blue	Good
**Solution B	Dark Green	Borderline
***Solution C	Yellow Green	Poor

* For Solution A, weigh accurately 1500 mg. of Na_2HPO_4 and dissolve in 20 ml. of distilled H_2O in a 100 ml. volumetric flask. Add 40 ml. of ethylene glycol and bring to volume with distilled H_2O.

** For Solution B, weigh accurately 900 mg. of Na_2HPO_4 and dissolve in 20 ml. of distilled H_2O in a 100 ml. volumetric flask. Add 40 ml. of ethylene glycol and bring to volume with distilled H_2O.

*** For Solution C, weight accurately 425 mg. of Na_2HPO_4 and dissolve in 20 ml. of distilled H_2O in a 100 ml. volumetric flask. Add 40 ml. of ethylene glycol and bring to volume with distilled H_2O.

The kit shall have the other reagent pad on the test strip react to various concentrations of antifreeze protection by changing to a color that is indicative of its freezing point level within a range that includes -60 °F to +10 °F.

Workmanship. The test strip shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). Each test kit containing fifty test strips, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement.

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Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Inspections and Testing. Inspection and testing shall be conducted in accordance with commercial practice to determine compliance with specification requirements.

Sampling for inspection and testing. Sampling for inspection and testing shall be conducted in accordance with commercial practice.

Certification of Commercial Item. The contractor shall certify that the product offered to the Government meets the salient characteristics of this description and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices and in the same product as the manufacturer's current product offered by the contractor in the commercial market place for at least one year preceding the solicitation. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Small business.

A small business that cannot meet the certification-of-commercial-item requirement of this document, may make an offer under this solicitation; and, if such small business is the apparent successful offeror, the Government may require the small business to submit a sample Test Kit, Test Strips and Color Chart, Antifreeze, Freeze Point and Corrosion. Any such Test Kit may be evaluated for this solicitation if time to contract award permits. Such evaluation would be to determine conformance to all requirements of this commercial item description and to determine equivalency to Antifreeze Test Kits available in the commercial marketplace in the areas of durability, human factors engineering, and safety. It should be noted that samples are not required unless specifically requested by the contracting officer. Note. The Government may elect not to evaluate a sample from an apparent low offeror for award for the solicitation if a determination of urgency is made by the contracting officer. The sample would then be evaluated prior to future solicitation for comparable items.

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Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document had been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and contents thereof under normal shipping conditions, handling. etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities, labeling, and marking shall be as specified in the contract or order.

The following National Stock Number is covered by this document:

6630-01-011-5039

Ordering data. (Expiration dating period required, intermediate/exterior package quantities, labeling, and marking must be specified in the contract or order).

MILITARY INTERESTS:

Custodian

Army - ME
Navy - MS
Air Force - 99

PREPARING ACTIVITY:

Army - ME

Project 6630-0483

Review activity

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITY:

VA-OSS

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